

Only One Event in Cox Model

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1 Introduction

Let's consider the Cox model and there are **No tied** observations. And define the distinct times of failure (or censoring) $\tau_1 < \tau_2 < \dots < \tau_n$. Denote

$$R_j = \{i | U_i \geq \tau_j\} = \text{risk set at } \tau_j,$$

where $U_i = \mathbf{min}(T_i, C_i)$ is the observed time and

$$Z_{(j)} = \text{value of } Z \text{ for the subject who fails at } \tau_j.$$

The partial likelihood $L_1(\beta)$ is defined as

$$L_1(\beta) \triangleq \prod_j \left\{ \frac{e^{\beta Z_{(j)}}}{\sum_{l \in R_j} e^{\beta Z_l}} \right\} \quad (1)$$

In this notes, we will consider the clinical trial scenario. So $Z_i \in \{0, 1\}$ for $i = 1, \dots, n$. $Z_i = 0$ denotes control group and $Z_i = 1$ denotes the treatment group. And we will consider the case when there's only one event observed.

2 Only One Event Observed

If there's only one event observed, then (1) can be simplified as

$$L_1(\beta) = \frac{e^{\beta Z_j}}{\sum_{l \in R_j} e^{\beta Z_l}} = \frac{e^{\beta Z_j}}{n_{trt} e^{\beta \times 1} + n_{ctrl} e^{\beta \times 0}} = \frac{e^{\beta Z_j}}{n_{trt} e^{\beta} + n_{ctrl}}, \quad (2)$$

where n_{trt} is the number of observations in R_j that belongs to treatment group and n_{ctrl} is the number of observations in R_j that belongs to the control group.

To maximize (2), we can see

$$L_1(\beta) = \begin{cases} \frac{e^\beta}{n_{trt}e^\beta + n_{ctrl}} = \frac{1}{n_{trt}} - \frac{n_{ctrl}}{n_{trt}} \frac{1}{n_{trt}e^\beta + n_{ctrl}} & \text{if } Z_j = 1 \\ \frac{1}{n_{trt}e^\beta + n_{ctrl}} & \text{if } Z_j = 0 \end{cases}$$

If $Z_j = 1$, (2) is maximized at $\beta = \infty$. If $Z_j = 0$, (2) is maximized at $\beta = -\infty$.

3 Only One Event in All Strata

Consider the stratified analysis scenario, where $k = 1, 2, \dots, K$ denotes the K strata levels. And the partial likelihood in each stratum is denoted as $L_1^{(k)}(\beta)$. The 'full' (in the sense of across all strata) partial likelihood is

$$L_1(\beta) = \prod_{k=1}^K L_1^{(k)}(\beta). \quad (3)$$

And if we further assume there's only one event in each and every strata level, then (3) becomes

$$L_1(\beta) = \prod_{k=1}^K \frac{e^{\beta Z_{j,k}}}{n_{trt,k}e^\beta + n_{ctrl,k}}$$

Without loss of generality, assume for $k = 1, 2, \dots, m$ the event happened in treatment group. For $k = m+1, m+2, \dots, m+n$ the event happened in control group. (Here the notation implies $K = m+n$.) Then the partial likelihood becomes

$$L_1(\beta) = e^{m\beta} \prod_{k=1}^{m+n} \frac{1}{n_{trt,k}e^\beta + n_{ctrl,k}}$$

And the partial log-likelihood is

$$l_1(\beta) = \text{Log} L_1(\beta) = m\beta - \sum_{k=1}^{m+n} \log(n_{trt,k}e^\beta + n_{ctrl,k})$$

Take derivative and we have

$$\frac{\partial l_1(\beta)}{\partial \beta} = m - \sum_{k=1}^{m+n} \frac{n_{trt,k}e^\beta}{n_{trt,k}e^\beta + n_{ctrl,k}} \quad (4)$$

We further assume $n_{trt,k} = pn_{ctrl,k}$ for $k = 1, \dots, K$, which means the treatment-control allocation is identical in each stratum. (This won't be **exactly** true in real-life but will hold on expectation. And I believe this is good enough for derivation.) Then (4) becomes

$$\frac{\partial l_1(\beta)}{\partial \beta} = m - \sum_{k=1}^{m+n} \frac{pe^\beta}{pe^\beta + 1} = m - (m+n) \frac{pe^\beta}{pe^\beta + 1}$$

Setting this to zero gives us

$$\hat{\beta} = \log m - \log(np) \quad \text{and} \quad e^{\hat{\beta}} = \frac{m}{np}$$

From this we can see, if for all strata levels, there's only one event in each of them, the estimated β and hazard ratio e^β depends on number of events happened in treatment and control group, and the allocation ratio between these two groups.

References